Zhenkai Zhang (Ken)

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EDUCATIONAL BACKGROUND

The Hong Kong Polytechnic University Starting Sep 2026

MSc in Data Science & Analytics

Nanjing Tech University Sep 2022 - Jun 2026

BSc in Data Science and Big Data Technology | 3.9/4.0

RESEARCH INTERESTS

Efficient AI; model reuse and routing; generative AI; multimodal agents

SELECTED PUBLICATIONS

Zhang, Z., Gao, Y., & Chen, D. (2025). Exploring and Enhancing Advanced MoE Models: From DeepSpeed-MoE to DeepSeek-V3. 2025 IEEE 6th International Seminar on Artificial Intelligence, Networking and Information Technology (AINIT). First author. **DOI**

Zhang, Z., Ma, T., Yao, Y., et al. (2025). Predicting Olympic Medal Performance for 2028: Machine Learning Models and the Impact of Host and Coaching Effects. Applied Sciences, 15(14), 7793. First author. **DOI**

Xia, W., Gao, Y., **Zhang, Z.**, et al. (2025). Behavior Prediction of Connections in Eco-Designed Thin-Walled Steel-Ply-Bamboo Structures Based on Machine Learning for Mechanical Properties. Sustainability, 17(15), 6753. Third author. **DOI**

SELECTED PROJECTS

RouterBench-Mini: Cost-Aware Model Reuse for Multimodal Agents Jul 2026

Independent Research Project | ***GitHub***

- Designed and implemented a 4,000-example cost-aware routing benchmark for Qwen3.5-35B-A3B and Qwen3.5-397B-A17B, using a 3,200-example development set and an 800-example frozen held-out test set spanning text, vision, and tool-use tasks.
- Evaluated frozen task-aware, learned quality-gap, and post-response Reflection routing using five-fold out-of-fold threshold selection, comprehensive overlap audits, feature ablations, matched random/oracle controls, paired bootstrap analysis, and failure analysis.
- On the frozen 800-example test set, Task-Aware trailed Always Strong by 0.63 percentage points while reducing average cost by 13.94% and latency by 14.01%, using the Strong model on 69.88% of tasks; released frozen configs, per-example predictions, reproducible scripts, protocol tests, and CI.

Finance-Agent: Multi-Agent Financial Analysis System Oct 2025 - Jan 2026

Engineering Project | ***GitHub***

- Designed and implemented an MCP-based multi-agent investment advisory system for structured analysis and recommendation generation across 4,000+ China A-share stocks, addressing data inconsistency and incomplete analytical coverage in general-purpose LLM systems.
- Collected and annotated approximately 100,000 financial news articles and fine-tuned Qwen3-2.5B with LoRA for sentiment classification and risk prediction, achieving test accuracies of 91% and 88%, respectively.
- Standardized financial APIs as MCP tools with a unified schema, achieving a 98% tool-call success rate and approximately 99% data consistency.
- Built four specialized agents for fundamental, technical, valuation, and news analysis using LangGraph and ReAct, together with a synthesis agent for generating structured Markdown reports.

TECHNICAL SKILLS

Python, PyTorch, Transformers, scikit-learn, PEFT/LoRA, LangGraph, MCP, Git, Linux, pytest

HONORS

Meritorious Winner (Problem F), Mathematical Contest in Modeling (MCM) Feb 2026